Heesang Yoo, Ph.D.

Personal Information

Nationality

South Korea

Address

307, 12 Rue du Dr Jean d'Argelos, 13100 Aix-en-Provence, France

Phone

(KR) +82-10-2991-8510

Email

yhs0365@gmail.com

Skills & Experience

Programming Language

- MATLAB, C++, CUDA
- Python

Research Expertise

- SPH, PD, DEM, FEM
- Multiphysics coupling
- HPC, GPU

Commercial Tools

- ANSYS Fluent
- COMSOL

System code

- MARS
- CATHARE

Open-source

- Basilisk (CFD)
- OpenFOAM (CFD)
- Modelica (system)
- C3PO (code coupling)

Others

- Paraview (visualization)
- Linux

Language

Korean (Native)

English (Advanced)

French (Basic)

EDUCATION

Ph.D. Seoul National University (SNU), Korea *2024*

Dept. of Nuclear Engineering (Thermal-Hydraulics)

Advisor Prof. Eung Soo Kim

Dissertation Multiphase ALE-SPH for Versatile Approaches on Highly Viscous Corium Simulation

B.E. Seoul National University (SNU), Korea *2017*

Dept. of Nuclear Engineering

WORK EXPERIENCE

Postdoc. French Alternative Energies and Atomic Energy Commission (CEA) Cadarache, France 2024.11 ~ ***present***

Aff. CEA/DES/IRESNE/SESI/LEMS

(Laboratory of Systems Studies and Modeling)

Advisor. Dr. Claire Vaglio-Gaudard

- Multiphysics Coupling of APOLLO3 and CATHARE3 via the C3PO Platform for Analysis of CABRI Power Transients

Postdoc. Seoul National University (SNU), Korea 2024.3 ~ 2024.10

Aff. Nuclear Research Institute for Future Technology and Policy.

Advisor Prof. Eung Soo Kim

RESEARCH INTERESTS

- **Development of High-Fidelity Multiphysics Simulation Tools**

Development and application of GPU-accelerated particle-based solvers (SPH/PD/DEM) for large-scale multiphysics and multiphase thermal-fluid phenomena.

- **Advanced Computational Methods for Coupled Multiphysics Systems**

Development of hybrid meshless–mesh-based numerical methods and scalable HPC frameworks for strongly coupled fluid–structure–thermal interactions.

- **Multi-Scale and System-Level Energy Systems Analysis**

Multi-scale modeling and integrated system-level analysis of advanced energy infrastructures, including SMR systems and high-density data center cooling.

- **Advanced Thermal Management for High-Power Applications**

Design and optimization of next-generation two-phase cooling technologies using high-fidelity simulations and thermos-economic assessment.

HONORS

- **Songwon–Kim Young-Hwan Scholarship** (2014–2018), Seoul National University, Korea.
- **Best Poster Award (First Author, Presenter)**, Transactions of the Korean Nuclear Society (KNS) Autumn Meeting, Yeosu, Korea, 2018.
- **Best Paper Award (Co-author)**, Transactions of the Korean Nuclear Society (KNS) Autumn Meeting, Gyeongju, Korea, 2023.
- **Best Paper Award (First Author, Presenter)**, Transactions of the Korean Nuclear Society (KNS) Spring Meeting, Jeju, Korea, 2024.
- **Best Paper Award (Co-author)**, 21st International Topical Meeting on Nuclear Reactor Thermal Hydraulics (NURETH-21), Busan, Korea, 2025.

MAJOR RESEARCH ACHEIVEMENTS

1. **[Journal Paper]** **Yoo, H. S,** & Kim, E. S.* (2024). Numerical investigation of Spreading Phenomena of Molten Corium using EISPH method. *Nuclear Engineering and Technology, 57(4), 103301.* **(2023-JCR-IF: 2.6, Nuclear Science & Technology Q1(7/41))**
   - **Role: First author**; Developed and applied the EISPH framework for severe accident analysis, performed large-scale simulations of corium spreading, and wrote the manuscript.

2. **[Journal Paper]** **Yoo, H. S,**, et al. (2024). Comparative study of WCSPH, EISPH and Explicit Incompressible-Compressible SPH (EICSPH) for Multi-Phase Flow with High Density Difference. *Journal of Computational Physics*, 112930. **(2024-JCR-IF: 3.8, Physics & Mathematical Q1(2/61))**
   - **Role: First author**; Designed comparative framework, implemented three SPH solvers, evaluated multiphase benchmarks with high density ratios, and led manuscript writing.

3. **[Journal Paper]** **Yoo, H. S**, & Kim, E. S.* (2025). Coupled EISPH–OSPD Framework for Modeling Hydroelastic Response and Brittle Fracture. Computational Mechanics, 1-35. **(2024-JCR-IF: 3.8, Interdisciplinary Applications & Mathematical Q1(16/136))**
   - **Role: First author**; Developed EISPH-OSPD scheme, optimized GPU parallelization, and demonstrated efficiency for various hydrodynamic problems with elastic-brittle deformation.

4. **[Journal Paper]** **Yoo, H. S,** et al. (2023). A simple Eulerian–Lagrangian weakly compressible smoothed particle hydrodynamics method for fluid flow and heat transfer. *International Journal for Numerical Methods in Engineering*, *124*(4), 928-958. **(2024-JCR-IF: 2.9, Interdisciplinary Applications & Mathematical Q1(26/136))** **(*Cover)**
   - **Role: First author**; Proposed a new Eulerian–Lagrangian WCSPH scheme, implemented the solver, conducted benchmark validations, and prepared the manuscript.

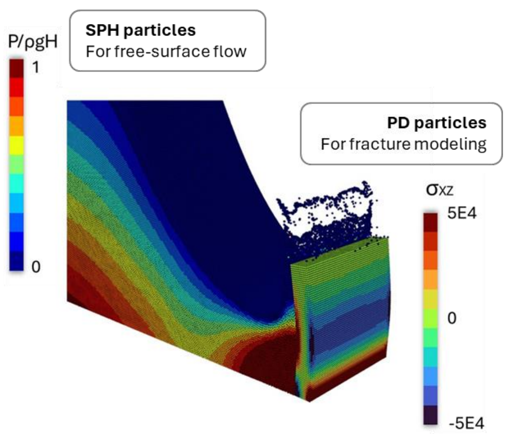
5. **[Journal Paper]** Kim, S. J, Yong, H, J., **Yoo, H. S.*** & Park, I. W,* (2026). From Water to Dielectric Fluids: Pool Boiling Research under Pressure and Subcooling for Next-Generation Cooling. *Renewable and Sustainable Energy Reviews* ***(Accepted)*** **(2024-JCR-IF: 16.3, Green & Sustainable Science & Technology Q1(26/136))**
   - **Role: Corresponding author** (led scientific direction); Conceptualization, methodological framework design, supervision, original draft preparation, and final manuscript revision

Research Focus Areas

1 High-Fidelity Multiphysics Thermal-Fluid Modeling

- Development of high-fidelity particle-based CFD methodologies for multiphase flow, phase-change phenomena, and conjugate heat transfer under extreme thermal-fluid conditions.
- Development of large-scale multiphysics simulation platforms for high-resolution transient analysis of strongly coupled nonlinear systems.
- Application of scalable computational frameworks to complex industrial and energy systems.

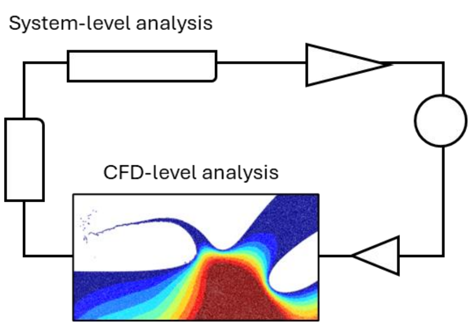
Related Publications: (J14) ~ (J2)



2 Advanced Computational Methods for Multiphysics Problem

- Development of coupled meshless/meshless and meshless/mesh-based numerical methods for fluid–structure–thermal interaction and strongly coupled multiphysics simulations.
- Design and implementation of GPU-accelerated high-performance computing (HPC) strategies for scalable large-scale transient analysis.
- Development of reduced-order modeling (ROM) and data-driven acceleration techniques for particle-based multiphysics simulations.

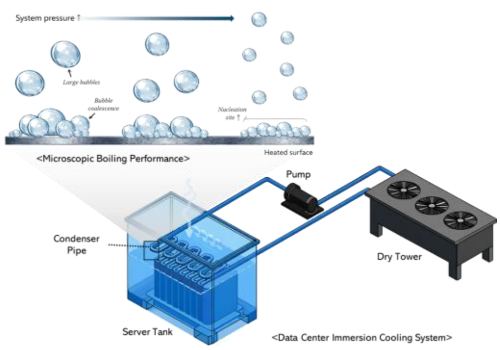
Related Publications: (J14), (J12), (J4)



3 Multi-Scale and System-Level Energy Systems Analysis

- System-level thermal-fluid analysis of large-scale energy infrastructures, including advanced nuclear reactor concepts and high-efficiency power cycles.
- Safety analysis and performance evaluation of nuclear reactors under steady-state and transient conditions
- Development of multi-scale simulation frameworks through coupling of system-level transient codes with three-dimensional CFD tools for integrated system analysis

Related Publications: (J18), (C15), (C9), (C8)



4 Advanced Thermal Management for High-Power Systems

- Development of next-generation two-phase cooling technologies for AI-driven data centers and high-power computing systems.
- Integration of high-fidelity numerical simulations, critical review of heat transfer performance, and thermos-economic analysis to enable optimized system-level thermal design.

Related Publications: (J15), (J16), (C14)

JOURNAL PUBLICATIONS

(J15) Kim, S. J, Yong, H, J., **Yoo, H. S.*** & Park, I. W,* (2026). From Water to Dielectric Fluids: Pool Boiling Research under Pressure and Subcooling for Next-Generation Cooling. *Renewable and Sustainable Energy Reviews* ***(Accepted)*** **(2024-JCR (2.4%, IF: 16.3))**

(J14) Ahn, Y. L., Choi, N. J., **Yoo, H. S.**, & Kim, E. S. (2026). Dynamic Load Balancing for Multi-GPU Smoothed Particle Hydrodynamics Using Two-Dimensional Staggered Domain Decomposition. *advanced in engineering software* ***(Accepted)*** **(2024-JCR (8.9%))**

(J13) Kim, T. H., **Yoo, H. S.**, Kim, J. W., & Kim, E. S. (2026). A Steady-State Eulerian Smoothed Particle Hydrodynamics (SPH) Approach for Incompressible Flow and Heat Transfer Using the Semi-Implicit Method for Pressure-Linked Equations (SIMPLE) Algorithm. *International Journal for Numerical Methods in Engineering*, *127*(1), e70255.

(J12) **Yoo, H. S.**, & Kim, E. S.* (2025). Coupled EISPH–OSPD framework for modeling hydroelastic response and brittle fracture. *Computational Mechanics, 1-35.* **(2024-JCR (11.4%))**

(J11) **Yoo, H. S.** & Kim, E. S.* (2025). Comparative Evaluation of Matrix-Free Implicit Viscosity Solvers for GPU-Accelerated SPH Simulation of Highly Viscous Fluids. *Computational Particle Mechanics, 1-24.*

(J10) Kim, J. H, **Yoo, H. S,** Jo, Y. B, Park, S. S, & Kim, E. S.* (2025). GPU-Parallelized SPH Solver for Accurate Simulation of Shaped Charge Jet Penetration in Concrete Structures. *International Journal of Fracture , 249(3), 52*,

(J9) **Yoo, H. S.** & Kim, E. S.* (2024). Numerical investigation of Spreading Phenomena of Molten Corium using EISPH method. *Nuclear Engineering and Technology, 57(4), 103301.*

(J8) Kim, J., **Yoo, H. S.**, Jo, Y. B., & Kim, E. S.* (2024). Simulation of Shockwave Propagation Characteristics in Nuclear Reactor Cavity during External Steam Explosion using Unified SPH. *Nuclear Engineering and Technology 57.4 (2025): 103274.*

(J7) Ahn, Y. L., **Yoo, H. S**., Choi, N. J., & Kim, E. S.* (2024). Dynamic Load Balancing of Multi-GPU Parallelization for VULCANO VE-U7 Corium Spreading Analysis using SOPHIA. *Nuclear Technology*, 1-21,

(J6) **Yoo, H. S.**, Jo, Y. B., & Kim, E. S.* (2024). Comparative study of WCSPH, EISPH and Explicit Incompressible-Compressible SPH (EICSPH) for Multi-Phase Flow with High Density Difference. *Journal of Computational Physics*, 112930. **(2024-JCR (2.5%))**

(J5) **Yoo, H. S.**, Jo, Y. B., Kim, J. W., Kim, E. S.*, & Choi, T. S. (2023). A simple Eulerian–Lagrangian weakly compressible smoothed particle hydrodynamics method for fluid flow and heat transfer. *International Journal for Numerical Methods in Engineering*, *124*(4), 928-958. **(*Cover)**

(J4) Jo, Y. B., Park, S. H., **Yoo, H. S**., & Kim, E. S.* (2022). GPU-based SPH-DEM Method to Examine the Three-Phase Hydrodynamic Interactions between Multiphase Flow and Solid

Particles. *International Journal of Multiphase Flow*, *153*, 104125.

(J3) **Yoo, H. S.**, Choi, H. Y., Kim, T. H., & Kim, E. S.* (2022). Effect of wettability on the water entry of spherical projectiles: Numerical analysis using smoothed particle hydrodynamics. *AIP Advances*, *12*(3).

(J2) Park, S. H., Jo, Y. B., Ahn, Y., Choi, H. Y., Choi, T. S., Park, S. S., **Yoo, H. S**., Kim, J.H. & Kim, E. S.* (2020). Development of multi-GPU–based smoothed particle hydrodynamics code for nuclear thermal hydraulics and safety: potential and challenges. *Frontiers in Energy Research*, *8*, 86.

(J1) **Yoo, H. S.**, Yoo, S. H., & Kim, E. S.* (2019). Heat transfer enhancement in dry cask storage for nuclear spent fuel using additive high density inert gas. *Annals of Nuclear Energy*, *132*, 108-118.

- **In Preparation & Under Review**

(J18) **Yoo, H.S.**, Debove, J., Vaglio-Gaudard. C*., Garcia-Cervantes. E.-Y., Jaboulay. J.-C., & Labit. J.-M.. (2026). Implementation of APOLLO3–CATHARE3 Multiphysics Coupling on the C3PO Platform for CABRI Power Transient Studies. *Annals of Nuclear Energy* **(In preparation)**

(J17) Kim, J.W, **Yoo, H. S**, & Kim, E. S.* (2026). A coupled EISPH-TLSPH method for the fluid-structure interaction of hydroelasticity with quasi-brittle fracturee. **(Under Review)**

(J16) Kim, S. J, Song, S, M., Park, H, S., **Yoo, H, S.** & Park, I. W,* (2026). Effects of Surface Characteristics of Oxidized Zircaloy-4 on Pool Boiling Heat Transfer. International Communications in Heat and Mass Transfer **(Under Review – Major Revision)**

CONFEREENCE PRESENTATIONS

(C15) **Yoo, H. S.**, Debove, J., Vaglio-Gaudard. C*., Garcia-Cervantes. E.-Y., Jaboulay. J.-C., & Labit. J.-M.. (2026). Multiphysics Coupling of APOLLO3 and CATHARE3 via the C3PO Platform for Analysis of CABRI Power Transients, International Conference on the Physics of Reactors (PHYSOR 2026), Turin, Italy, April 19-23, 2026. **(*Accepted*)**

(C14) Kim, M. H., **Yoo, H. S.**, & Park, I. W. (2025) Numerical Analysis of Melting and Interfacial Flow in a Direct-Contact Latent Heat Storage System Reflecting the Marangoni Effect, *Korean Society for Fluid Machinery (KSFM)*, Jeju, Korea, December 4-5, 2025 (in Korean)

(C13) Kim, E. S.*, Ahn, Y., Kim, J.-W., Kim, J. H., Park, J., Jo, Y. B., & **Yoo, H. S.** (2025). Particle-Based Approaches to Multiphysics Simulation in Nuclear Safety, 21*th International Topical Meeting on Nuclear Reactor Thermal Hydraulics (NURETH 21)*, Busan, Korea, August 31 - September 5, 2025.

(C12) Kim, J. W., **Yoo, H. S.**, Lee, T. H., Ahn, Y., Kwon, M. C., & Kim, E. S.* (2025) A Smoothed Particle Hydrodynamics Model for Safety Assessment of the Ex-vessel Core Catcher under Reactor Vessel Failure, 21*th International Topical Meeting on Nuclear Reactor Thermal Hydraulics (NURETH 21)*, Busan, Korea, August 31 - September 5, 2025. **(Best Paper)**

(C11) Kim, J. W., **Yoo, H. S.**, Lee, T. H., Ahn, Y., Kwon, M. C., & Kim, E. S.* (2024) Numerical

Investigation on the Safety Assessment of an Ex-vessel Core Catcher under Large-break Reactor Vessel Failures Using Smoothed Particle Hydrodynamics, *The Eleventh Korea-Japan Symposium on Nuclear Thermal Hydraulics and Safety*, Seoul, Korea, November 10-13, 2024

(C10) **Yoo, H. S.**, & Kim, E. S.* (2024). Numerical Investigation of Corium Spreading Phenomena using Smoothed Particle Hydrodynamics, Transactions of the Korean Nuclear Society Spring Meeting, Jeju, Korea, May 9-10, 2024 (in Korean) **(Best Paper)**

(C9) Park, K. J., Kim, T. H., **Yoo, H. S**., Jo, Y. B., & Kim, E. S.* (2024). Dynamic Modeling for 20 kWe Heat Pipe Fission Battery with Dual Power Conversion System, Transactions of the Korean Nuclear Society Spring Meeting, Jeju, Korea, May 9-10, 2024 (in Korean)

(C8) Park, K. J., Kim, T. H., **Yoo, H. S**., Kim, E. S.*, & Jo, Y. B (2023). Dynamic Modeling of Free Piston Stirling Generator for Micro Nuclear Reactors, *Transactions of the Korean Nuclear Society Autumn Meeting* Gyeong-Ju, Korea, October 25-27, 2023 (in Korean) **(Best Paper)**

(C7) **Yoo, H. S.**, Jo, Y. B., & Kim, E. S.* (2022) GPU-accelerated Explicit Incompressible-Compressible SPH for multi-phase flow with large density difference, *Proceedings of the SPHERIC Workshop, Piazza Roma, Catania. Italy*

(C6) **Yoo, H. S.**, & Kim, E. S.* (2021) Development of an Eulerian–Lagrangian Coupling Method Using SPH, *Proceedings of the KSCFE Conference*, Seoul National University Siheung Campus, Korea. (in Korean)

(C5) **Yoo, H. S.**, & Kim, E. S.* (2021) Comparative Analysis of Recent Pressure-Noise Reduction Methods for Free-Surface Flow and Fluid–Structure Interaction Using SPH, *Proceedings of the KSOE Conference*, Busan, Korea. (in Korean)

(C4) **Yoo, H. S.**, Kim, T. H., & Kim, E. S.* (2021). Numerical Simulation on Spreading and Impact Behavior of a Single Droplet Using Smoothed Particle Hydrodynamics. *Transactions of the Korean Nuclear Society Virtual Autumn Meeting*, October 21-22, 2021 (in Korean)

(C3) **Yoo, H. S.**, Yoo, S. H., & Kim, E. S.* (2018) Heat Transfer Enhancement of Dry Cask Storage System Using Binary Helium and Heavy Inert Gas Mixture, *The Eleventh Korea-Japan Symposium on Nuclear Thermal Hydraulics and Safety*, Busan, Korea, November 18-21, 2018

(C2) **Yoo, H. S.**, Kim, E. S.*, & Yoo, S. H. (2018). Thermal Performance Enhancement of Dry Cask Storage System Using Helium-Based Binary Gaseous Mixture. *Transactions of the Korean Nuclear Society Autumn Meeting*, Yeosu, Korea, October 25-26, 2018 (in Korean) **(Best Poster)**

(C1) Song, M. S., **Yoo, H. S**., Kim, J. H., & Kim, E. S.* (2017). The investigation of flow characteristics in 3-pin wire-wrapped fuel bundle using laser-based measurement techniques. *17th International Topical Meeting on Nuclear Reactor Thermal Hydraulics (NURETH 17)*, Xi'an, China, September 3–8, 2017

SOFTWARE DEVELOPMENT

- **SOPHIA**. **a Lagrangian-based HPC-CFD code for versatile thermal-hydraulics and nuclear safety application**
  , NEA-1911 (https://www.oecd-nea.org/tools/abstract/detail/nea-1911/) PI: Kim, E. S.
  Co-Developers: Jo, Y. B., Park, S. H., Choi, H. Y., Choi, T. S., Park, S. S., **Yoo, H. S.**, Ahn, Y. L., Kim, J. H, Lee., T. H, Chae, H., Kim, J. W., Park, J. R., Kim, D. H.

PROFESSIONAL SERVICE

- **Technical Review:**
  The Korean Hydrogen & New Energy Society, Korean Nuclear Society,
  Ocean Engineering, Nuclear Engineering and Technology,
  Physics of Fluids

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **President**, SNU Global Outreaching & Education Club (2015)
  Organized academic exchange events and coordinated volunteer programs with international students.

TEACHING & SUPERVISION EXPERIENCE

- **Undergraduate Thesis Supervision** *Tae Hwan Kim* (B.S. thesis, 2020) – awarded *Best Thesis Prize*
- **Teaching Assistant**
  - Engineering of Energy Systems (2017)
  - High-Performance Particle Methods for Thermal–Hydraulic Analysis (2021)

LIST OF REFERENCES

1. **Prof. Eung Soo Kim** 🎓 ***(Dissertation Supervisor)***
   Professor, Department of Nuclear Engineering, Seoul National University
   Gwanak-gu, Gwanak-ro, Seoul, Republic of Korea
   🖂 **Email**: kes7741@snu.ac.kr
2. **Dr. Claire Vaglio-Gaudard** 🔬 *(Current Advisor)*
   Senior Expert & Research Engineer, CEA Cadarache
   Cadarache, 13108 Saint-Paul-lès-Durance, France
   🖂 **Email**: claire.vaglio-gaudard@cea.fr
3. **Prof. Il-Woong Park** 🔬 *(Collaborator)*
   Multiphase Flow and Thermal Engineering
   Department of Mechanical Engineering, Inha University
   🖂 **Email**: ilwoongpark@inha.ac.kr
4. **Prof. Qiao Wu** 🎓 *(Collaborator)*
   Director of NuScale Integral System Testing program
   School of Nuclear Science and Engineering, Oregon State University
   Corvallis, OR 97441, USA
   🖂 **Email**: Qiao.Wu@oregonstate.edu
5. **Dr. Nam-il Tak** 🔬 *(Dissertation Committee)*
   Principal Research, Department of High Temperature Reactor, KAERI
   111, Daedeok-daero 989beon-gil, Yuseong-gu, Daejeon, Republic of Korea
   🖂 **Email**: takni@kaeri.re.kr